

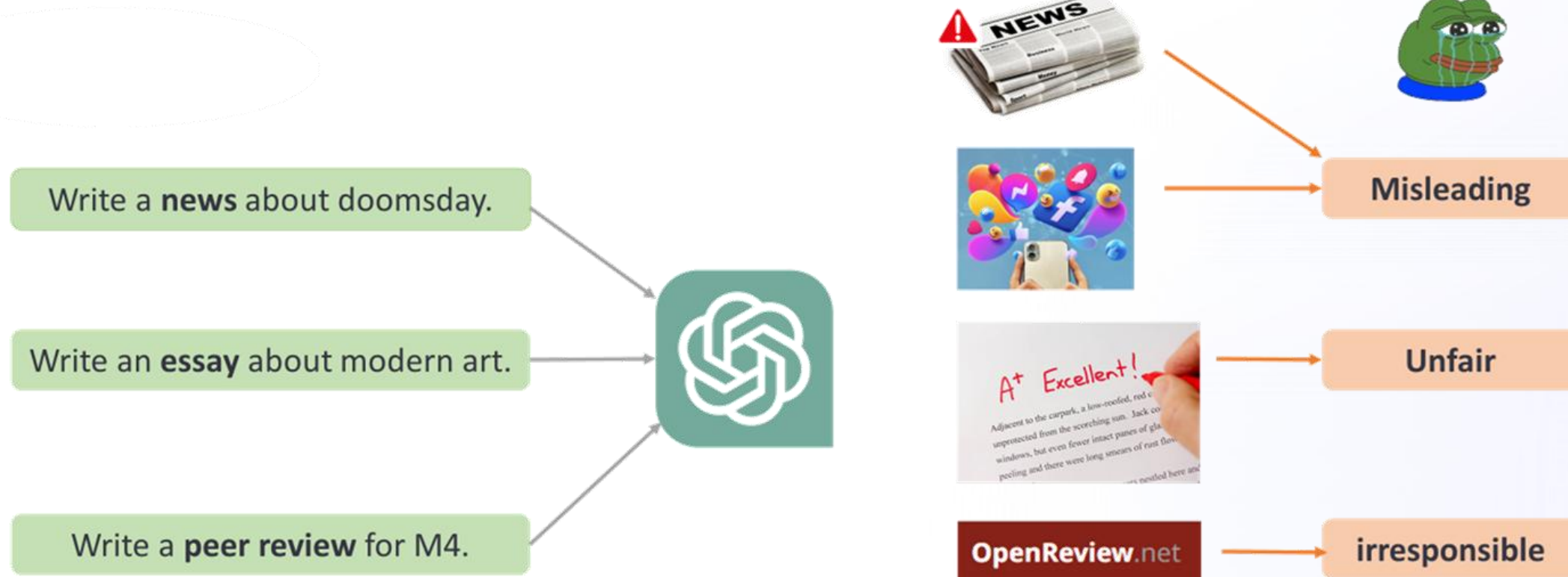
**SemEval-2024 Task 8:
Multidomain, Multimodel and Multilingual
Machine-Generated Text Detection**



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Problem



Misuse of machine-generated text is harmful.



Objective

- ❑ **Advance** the development of automated systems for **detecting MGT** (machine-generated text)
- ❑ **Reduce** instances of the potential **misuse of MGT** in a variety of fields such as journalism, education, academia



Outline

- Problem
- Objective
- **Tasks**
- Dataset
- Evaluation Metrics
- Task Organization
- Participant Submissions



Tasks

- ❑ **Subtask A: Human vs. Machine Classification:** classify a text as either produced by a human or generated by a machine.
- ✓ English track
- ✓ Multilingual track: English, Chinese, Arabic, Bulgarian, Russian, Indonesian, Urdu, Italian, German



Tasks

- ☐ Subtask A: Human vs. Machine Classification
- ☐ **Subtask B: Multi-Way Generator Detection:** pinpoint the exact source of a text.
- ✓ English track: determine whether text is originated from a human or a specific LLM (davinci-003, GPT-3.5, GPT-4, Cohere, Dolly-v2, or BLOOMz).



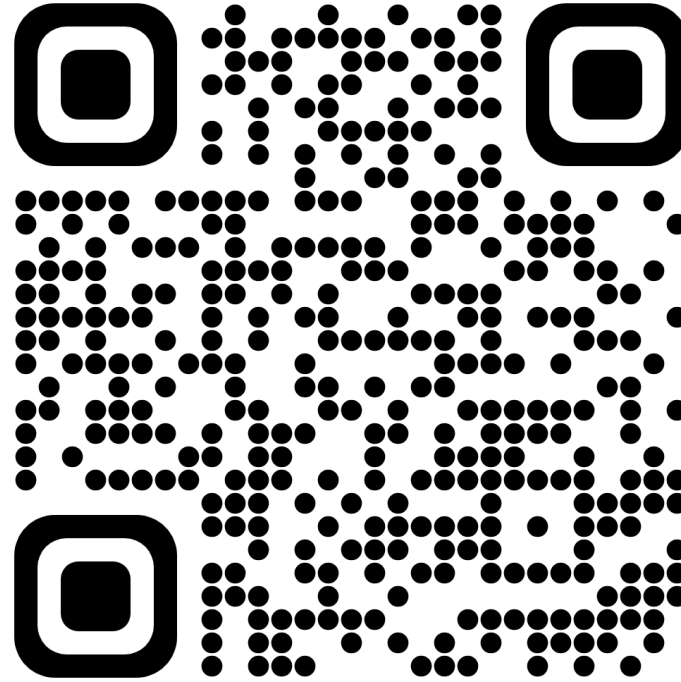
Tasks

- ☐ Subtask A: Human vs. Machine Classification
- ☐ Subtask B: Multi-Way Generator Detection
- ☒ **Subtask C: Change Point Detection**

- ✓ English track: estimate the authorship transition point from human to machine.



Dataset



M4

Best Resource Paper Award



EACL 2024



Dataset Overview

9 Languages

9 Generators

7 Domains



Dataset Overview

9 Languages

- ❖ English
- ❖ Chinese
- ❖ Arabic
- ❖ Bulgarian
- ❖ Russian
- ❖ Indonesian
- ❖ Urdu
- ❖ Italian
- ❖ German



Dataset Overview

9 Generators

- ✓ GPT-3 (davinci-003)
- ✓ GPT-3.5 / ChatGPT
- ✓ GPT-4
- ✓ Cohere
- ✓ Dolly-v2 7B
- ✓ BLOOMz 176B
- ✓ LLaMA 7B, 13B, 70B
- ✓ Italian LLaMA2 7B, 13B, 70B
- ✓ Jais



Dataset Overview

7 Domains

- ☐ News articles
- ☐ Scientific papers
- ☐ Paper reviews
- ☐ Wikipedia
- ☐ Wikihow
- ☐ Social media
- ☐ Student essays



Split	Source	davinci-003	ChatGPT	Cohere	Dolly-v2	BLOOMz	GPT-4	Machine	Human
Train	Wikipedia	3,000	2,995	2,336	2,702	—	—	11,033	14,497
	Wikihow	3,000	3,000	3,000	3,000	—	—	12,000	15,499
	Reddit	3,000	3,000	3,000	3,000	—	—	12,000	15,500
	arXiv	2,999	3,000	3,000	3,000	—	—	11,999	15,498
	PeerRead	2,344	2,344	2,342	2,344	—	—	9,374	2,357
Dev	Wikipedia	—	—	—	—	500	—	500	500
	Wikihow	—	—	—	—	500	—	500	500
	Reddit	—	—	—	—	500	—	500	500
	arXiv	—	—	—	—	500	—	500	500
	PeerRead	—	—	—	—	500	—	500	500
Test	Outfox	3,000	3,000	3,000	3,000	3,000	3,000	18,000	16,272

Dataset – Subtask A - Multilingual Track

Split	Language	davinci-003	ChatGPT	LLaMA2	Jais	Other	Machine	Human
Train	English	11,999	11,995	—	—	35,036	59,030	62,994
	Chinese	2,964	2,970	—	—	—	5,934	6,000
	Urdu	—	2,899	—	—	—	2,899	3,000
	Bulgarian	3,000	3,000	—	—	—	6,000	6,000
	Indonesian	—	3,000	—	—	—	3,000	3,000
Dev	Russian	500	500	—	—	—	1,000	1,000
	Arabic	—	500	—	—	—	500	500
	German	—	500	—	—	—	500	500
Test	English	3,000	3,000	—	—	9,000	15,000	13,200
	Arabic	—	1,000	—	100	—	1,100	1,000
	German	—	3,000	—	—	—	3,000	3,000
	Italian	—	—	3,000	—	—	3,000	3,000



Dataset – Subtask B

Split	Source	davinci-003	ChatGPT	Cohere	Dolly-v2	BLOOMz	Human
Train	Wikipedia	3,000	2,995	2,336	2,702	2,999	3,000
	Wikihow	3,000	3,000	3,000	3,000	3,000	2,995
	Reddit	3,000	3,000	3,000	3,000	2,999	3,000
	arXiv	2,999	3,000	3,000	3,000	3,000	2,998
Dev	PeerRead	500	500	500	500	500	500
Test	Outfox	3,000	3,000	3,000	3,000	3,000	3,000



Dataset – Subtask C

Domain	Generator	Train	Dev	Test	Total
PeerRead	ChatGPT	3,649 (232)	505 (23)	1,522 (89)	5,676 (344)
	LLaMA-2-7B*	3,649 (5)	505 (0)	1,035 (1)	5,189 (6)
	LLaMA-2-7B	3,649 (227)	505 (24)	1,522 (67)	5,676 (318)
	LLaMA-2-13B	3,649 (192)	505 (24)	1,522 (84)	5,676 (300)
	LLaMA-2-70B	3,649 (240)	505 (21)	1,522 (88)	5,676 (349)
OUTFOX	GPT-4	—	—	1,000 (10)	1,000 (10)
	LLaMA2-7B	—	—	1,000 (8)	1,000 (8)
	LLaMA2-13B	—	—	1,000 (5)	1,000 (5)
	LLaMA2-70B	—	—	1,000 (19)	1,000 (19)
Total	all	18,245	2,525	11,123	31,893



Evaluation Metrics

- ❑ Subtask A: Binary Classification
 - Balanced dataset → Accuracy
- ❑ Subtask B: Multiclass Classification
 - Balanced dataset → Accuracy
- ❑ Subtask C: Change point Detection
 - Mean Absolute Error (MAE)



Outline

- Problem
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- Tasks
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- **Task Organization**
- Participant Submissions



Task Organization: Platforms & Tools

- ❑ Task Details: **GitHub**
- ❑ Communications:
 - **Google Groups** (76 emails)
 - **CodaLabs** (53 posts)
- ❑ Submissions: **CodaLab**
- ❑ Leaderboard: **Google Sheet**
- ❑ Data Sharing: **Google Drive**
- ❑ System Descriptions + BibTex:
 - **Google Doc**
 - **Google Sheet**



Task Organization: Phases

- ☐ **Development phase**
 - ☐ No gold labels for dev set
 - ☐ Unlimited submissions
 - ☐ Best results shown in live leaderboard



Task Organization: Phases

- ☐ Development phase
- ☐ **Testing phase**
 - ☐ Timeframe: 3 weeks
 - ☐ Unlimited submissions BUT
 - ☐ Results are NOT displayed until AFTER the deadline
 - ☐ Only last submission is considered



Task Organization: Phases

- ☐ Development phase
- ☐ Testing phase
- ☐ **Test data – Surprise languages, domains, and models**
 - ☐ Subtask A
 - ☐ 2x additional languages (German and Italian) – Multilingual track
 - ☐ 1x new generator: GPT-4 – Monolingual track
 - ☐ Subtask B
 - ☐ 1x new domain: student essays
 - ☐ Subtask C
 - ☐ 2x new domains (PeerRead + OUTFOX)
 - ☐ 2x new generators (GPT-4 + LLaMa 2)





SEMEVAL-2024, TASK 8: MULTIGENERATOR, MULTIDOMAIN, AND MULTILINGUAL BLACK-BOX MACHINE-GENERATED TEXT DETECTION

316

PARTICIPANTS

1308

SUBMISSIONS

ORGANIZED BY: Aktsvigun

CURRENT PHASE ENDS: Never

CURRENT SERVER TIME: May 18, 2024 At 7:42 PM GMT+4

Docker image: [codalab/codalab-legacy:py39](#) 



CodaLab Review

- ✓ Convenient platform for user login management
- ✓ Allows users to make submissions
- ❖ A bit slow at showing scores to users after they make a submission
- ❖ This resulted in many participant complaints
- ❖ Scoring was eventually done manually
- ❖ Leaderboard was created on a Google sheet



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Submissions Stats

❑ Subtask A monolingual: **126**

Accuracy = **97.53**

❑ Subtask A multilingual: **59**

Accuracy = **95.99**

❑ Subtask B: **70**

Accuracy = **90.85**

❑ Subtask C: **30**

MAE = **15.68** ↓



Subtask A Approaches

Team Name	Ranking	small PLM	LLM	GPT	Fine-tuning	Zero-shot	Data augmentation	External Data
USTC-BUPT	1		✓		✓			
FI Group	2	✓			✓			
KInIT	3		✓		✓	✓		
L3i++	5		✓		✓			
QUST	6		✓		✓		✓	
AIpom	9		✓		✓			
SINAI	21		✓		✓			
Unibuc-NLP	22	✓			✓			
Werkzeug	30	✓			✓			
RUG-5	32	✓	✓		✓			
DUTh	33		✓		✓			
RUG-D	39		✓		✓			✓
MasonTigers	49	✓	✓		✓	✓		
TrustAI	55	✓			✓			

□ Monolingual

Team Name	Ranking	small PLM	LLM	GPT	fine-tuning	zero-shot	few-shot (k=?)	Data augmentation	External Data
Genaios	1		✓						
USTC-BUPT	2	✓			✓				
petkaz	12	✓			✓				
HU	17		✓		✓			✓	
TrustAI	20	✓			✓				
L3i++	25		✓		✓				
art-nat-HHU	26	✓			✓				
Unibuc - NLP	28	✓			✓			✓	
NewbieML	30	✓							
QUST	31		✓		✓			✓	
NootNoot	39	✓			✓				
Mast Kalandar	40		✓		✓				
I2C-Huelva	41		✓		✓				
Werkzeug	45	✓			✓				
NCL-UoR	50		✓		✓				
Sharif-MGTD	51		✓		✓	✓			
Collectivized Semantics	62	✓			✓				
SINAI	61		✓		✓				
MasonTigers	71	✓	✓		✓	✓			
DUTh	73		✓		✓				
surbhi	74		✓		✓				
KInIT	77		✓		✓	✓			
RUG-D	100		✓		✓				✓
RUG-5	101	✓	✓		✓				
RUG-3	114		✓	✓	✓				
Mashee	115		✓				2		
RUG-1	117	✓							

□ Multilingual



Subtask B Approaches

Team Name	Ranking	small PLM	LLM	GPT	fine-tuning	zero-shot	Data augmentation
AISPACE	1		✓		✓		✓
Unibuc - NLP	2	✓			✓		✓
USTC-BUPT	3		✓				
L3i++	6		✓		✓		
MLab	7				✓		
Werkzeug	8	✓			✓		
TrustAI	14	✓			✓		
MGTD4ADL	17				✓		✓
scalar	18	✓					✓
UMUT	23	✓			✓		
QUST	36		✓		✓		✓
MasonTigers	38	✓	✓		✓	✓	
RUG-5	41	✓	✓		✓		
RUG-D	44		✓		✓		
DUTh	49		✓		✓		
clulab-UofA	62		✓	✓	✓		✓



Subtask C Approaches

Team Name	Ranking	small PLM	LLM	LSTM (+) CNN	fine-tuning	Data augmentation	CRF layer
TM-TREK _{STC:1}	1	✓			✓		✓
Alpom _{STC:2}	2		✓		✓		
USTC-BUPT _{STC:3}	3	✓	✓		✓	✓	
RKadiyala _{STC:6}	6	✓			✓		✓
DeepPavlov _{STC:7}	7	✓			✓	✓	
RUG-5 _{STC:17}	17	✓	✓		✓		
TueCICL _{STC:22}	22			✓			
jelarson _{STC:25}	25						
MasonTigers _{STC:27}	27	✓					
Unibuc-NLP _{STC:28}	28			✓	✓		



Participants Paper Submissions

55 submissions, **48** accepted
87% acceptance rate



Organizers

- Yuxia Wang,[†]
- Jonibek Mansurov,[†]
- Petar Ivanov,[†]
- Jinyan Su,[†]
- Artem Shelmanov,[†]
- Akim Tsvigun,[†]
- Osama Mohammed Afzal,[†]
- Tarek Mahmoud,[†]
- Giovanni Puccetti,[§]
- Thomas Arnold,[¶]
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Thank you!

